

Test Report: Supporting Structure Stability

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Project: Substrate Mixer and Portioning Device

Component: Supporting Structure

Prototype Version: Final Application

1 PART I — TEST PLAN

1.1 Research Hypothesis

The supporting structure for the oil barrel is hypothesized to provide sufficient static and dynamic stability. It must remain stationary and resist tilting under operational conditions, including high dead weight, a substrate load exceeding 100l, and the torques generated during the agitation (stirring) process.

1.2 Test Objective

To verify the hypothesis, the following evaluations will be conducted:

- **Manual Vibration Test:** Visual inspection for structural deformation or displacement under oscillating loads.
- **Determination of Tilting Moments:** Quantification of the force required to tilt the assembly along both the longitudinal and radial axes.

1.3 Measurement Methodology

Setup: An empty oil barrel is mounted within the supporting structure (without agitator or substrate).

Instruments: Analog force gauge.

Procedure: Force is applied at specific points until the structure begins to lift from its base (tipping point). The tilting moment (M) is calculated as $M = F \cdot l$ (Bender & Göhlich, 2020).

Force Application Points:

- **Radial:** Center of the upper edge of the rear panel.
- **Longitudinal:** Center of the upper rim of the mounted oil barrel.

1.4 Success Criteria

The hypothesis is validated if the stability factor (S) is greater than 1.0 for all axes (Bender & Göhlich, 2020):

$$S = \frac{M_{\text{restoring}}}{M_{\text{tilting}}} > 1 \quad (1)$$

- **Restoring Moment ($M_{\text{restoring}}$):** Total weight (barrel + structure) $\times$ distance to the tipping edge.
- **Tilting Moment (M_{tilting}):** Measured force at the tipping point $\times$ lever arm (distance to the tipping edge).

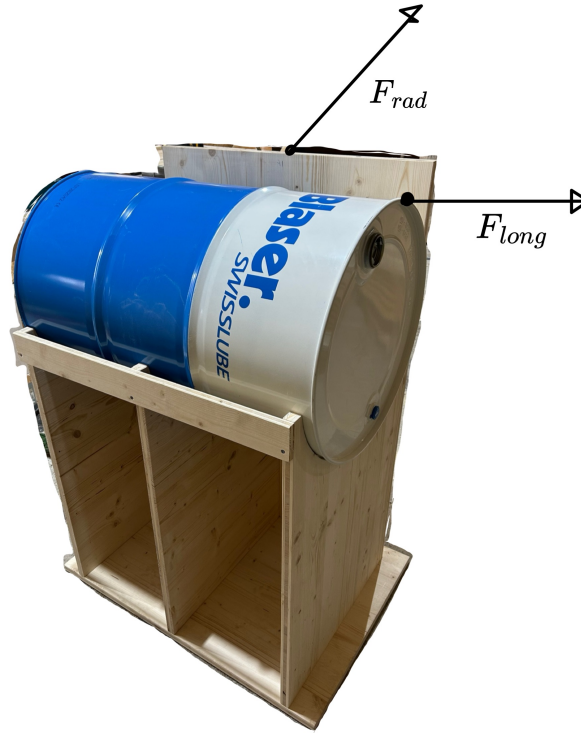


Figure 1: Picture of supporting Structure with oil barrel. Force application points in longitudinal (F_{long}) and in radial (F_{rad}) direction marked.

2 PART II — TEST EXECUTION DATA

2.1 Mass and Geometry Data

The mass of the structure was calculated based on the volume of spruce wood ($\rho \approx 470 \text{ kg/m}^3$) and the barrel weight.

Parameter	Value
Mass of Oil Barrel (m_{barrel})	$\approx 17 \text{ kg}$
Mass of Structure (m_{frame})	$\approx 42 \text{ kg}$
Total Weight (F_G)	$\approx 590 \text{ N}$
Distance to Tipping Edge (Longitudinal)	0.50 m
Distance to Tipping Edge (Radial)	0.35 m

2.2 Tilting Moment Measurements

Axis	Height (l)	Force (F)	Tilting Moment (M_t)
Longitudinal	1.28 m	180 N	230 Nm
Radial	1.30 m	130 N	170 Nm

2.3 Stability Calculation

Axis	$M_{\text{restoring}}$	M_{tilting}	Stability Factor (S)
Longitudinal	295 Nm	230 Nm	1.28
Radial	206 Nm	170 Nm	1.21

3 PART III — TEST REVIEW

3.1 Observations

- **Manual Vibration Test:** The structure exhibited high stiffness. No deformations, shifting, or structural resonances were observed during manual agitation.
- **Build Quality:** The assembly provides a robust and reliable impression suitable for the intended load.

3.2 Data Interpretation

Measurements confirm that restoring moments significantly exceed tilting moments in both critical axes. Stability factors ($S_{\text{long}} = 1.28$, $S_{\text{rad}} = 1.21$) meet the safety requirements for the empty state.

Note: This test should be considered as a rough assessment. A detailed analysis has been deliberately omitted as this would go beyond the scope of this work. The Center of Gravity (CoG) was approximated based on the geometric distribution of mass. Dynamic influences of a rotating agitator and resulting fluid dynamics (sloshing effects) were not quantitatively assessed.

3.3 Knowledge Gained

The supporting structure is verified as stable. The weight distribution and base geometry are sufficient to handle operational forces without additional anchoring.

3.4 Hypothesis Evaluation

☒ Confirmed

☐ Partially confirmed

☐ Rejected

Explanation: All success criteria were met. Stability factors are consistently > 1.0 .

3.5 Next Development Steps

No further design iterations are required. The development of the Vessel and Supporting Structure is hereby successfully concluded.

References

Bender, B., & Göhlich, D. (2020). *Doppel Taschenbuch für den Maschinenbau 1: Grundlagen und Tabellen* (26th ed.). Springer. <https://doi.org/https://doi.org/10.1007/978-3-662-59711-8>